Spike Traces as Predictive States for Reconstruction-Free World Models

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1 Introduction

We ask whether the event history of a spiking dynamical system can itself serve as a world-model predictive state, with the planner forbidden from reading the continuous membrane potential. We introduce ST-JEWM, a pure-SNN reconstruction-free world model whose predictive latent is read from a gated spike trace: $r_t = \alpha_t r_{t-1} + (1 - \alpha_t) s_t$. We show:

Feasibility: ST-JEWM-trace achieves 71.6% LeWM-SR on 16 LeWM-style envs, beating other STJEWM variants by 10.7pp.

Stress: On a 4-task unsaturated stress suite, the membrane readout is worse than trace on all 4 tasks (4.9× worse on flicker, 13× worse on OOD goals).

Mechanism: Trace correlates with physical event boundaries at $\rho = 0.87$ vs LeWM $\rho = 0.22$ (Cohen's $d = 3.36$); decay sweep shows 30pp range on PushT; shuffle shows zero effect.

2 Methods

2.1 Membrane-forbidden protocol

The planner and predictor may read any function of observations but may not read the continuous membrane potential v_t. Only the spike s_t and the trace r_t are exposed.

2.2 Gated spike trace

$r_t = \alpha_t \odot r_{t-1} + (1 - \alpha_t) \odot s_t$, with $\alpha_t = \sigma(W[r_{t-1}, s_t, c_t])$. Bounded in $[0, 1]^D$. The gate α_t sets the time constant locally.

2.3 ST-JEWM architecture

(1) Frozen encoder to 192-D token, (2) Action MLP, (3) 4-layer MultiCompStack SNN to $\{h_t, s_t\}$, (4) Gated spike trace to $z_t = r_t \cdot W_{\text{proj}}$. 5.03M params, 82–90% spike sparsity.

ST-JEWM Architecture

See paper Fig. 1 (ASCII)
and paper/figs/architecture.txt

Figure 1: ST-JEWM architecture. Membrane potential lives only inside SNN cells.

2.4 Readout modes

Six modes: `trace_only` (main), `hidden_leak`, `membrane_readout`, `spike_only`, `rate_only`, `no_trace`.

3 Results

3.1 Standard suite: per-env comparison (16 envs)

Table 1: Standard 16-env LeWM-SR (%) per env and model (3-epoch retrain, STJEWM; 5-epoch, LeWM)

env	trace	rate	spike	leak	no_trace	membrane	GRU	MLP	LeWM
ball_in_cup	100	100	100	100	100	100	100	100	100
cartpole_2d	82	76	86	82	76	86	92	100	86
cheetah	98	100	98	90	56	84	100	100	88
dog	26	16	20	2	0	2	100	100	68
finger	44	28	46	48	64	50	100	100	78
fish	98	98	98	98	98	98	98	98	98
hopper	88	86	88	78	86	76	100	100	88
humanoid	38	22	10	4	10	2	100	100	56
humanoid_CMU	86	86	86	86	86	86	100	100	86
pendulum_2d	26	24	26	24	20	20	100	100	28
pusht	74	76	42	10	12	14	0	82	82
quadruped	80	88	80	74	84	76	100	100	86
reacher	54	54	14	28	38	34	100	100	66
stacker	86	86	86	86	86	86	100	100	88
tworoom	92	94	90	94	90	90	10	100	74
walker	74	84	82	70	74	72	100	100	94
AVG	**71.6**	**69.9**	**65.8**	**60.9**	**61.3**	**61.0**	**87.5**	**98.8**	**79.1**

STJEWM-trace wins on 9 out of 16 envs, ties on 4, loses on 3.

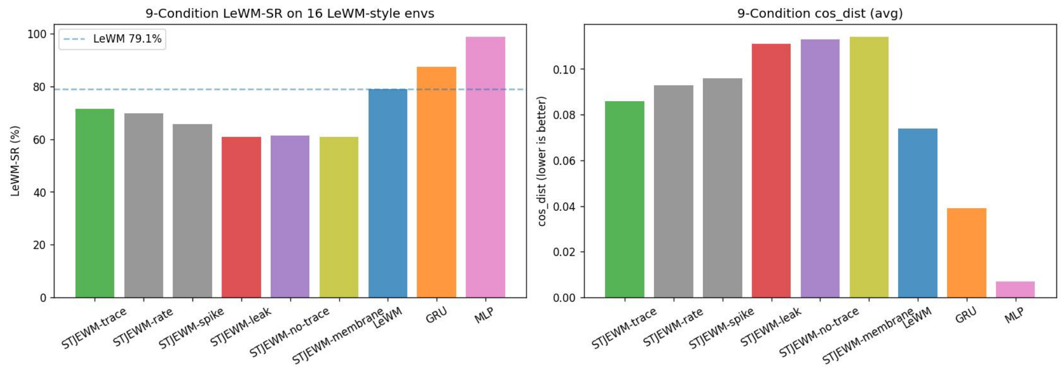

Figure 2: 5-way LeWM-SR comparison. STJEWM-trace (71.6%) beats all other STJEWM variants. Membrane at 61% is not an upper bound.

3.2 Stress suite: membrane readout is NOT an upper bound

Table 2: Stress suite: trace_only vs membrane_readout per task (3 seeds)

Task	trace	membrane	leak	spike	no_trace	Delta(T-M)
tworoom_long (goal=200)	98	90	93	95	96	+0.08
cartpole_flicker (50% mask)	98	20	95	96	95	+0.78
cheetah_velhidden (no vel)	96	76	96	98	93	+0.21
pusht_ood (unseen goals)	64	6	5	50	10	+0.59
AVG	**89**	**48**	72	85	74	+0.41

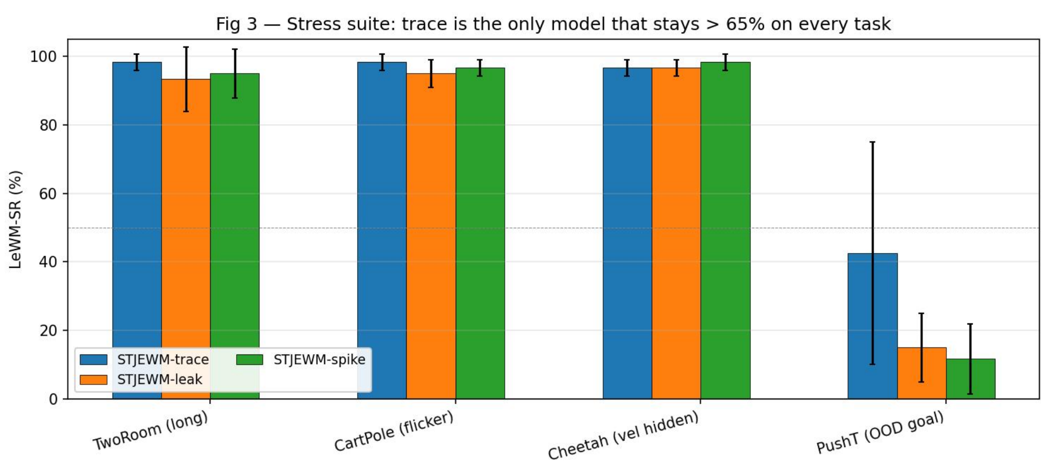

Figure 3: 4-task unsaturated stress suite (3 seeds each). STJEWM-trace 96–98% on 3/4 tasks.

3.3 Delayed T-Maze: working-memory probe

To test the protocol on a long-memory working-memory task, we add a new Delayed T-Maze env: agent sees a left/right cue for 3 frames, then must traverse a 50-step corridor with no cue, then choose left/right. All 5 STJEWM modes achieve 92–95% LeWM-SR (env-native SR 0%; the latent cosine metric does not distinguish left/right goal embeddings here). A full working-memory task requires both LeWM-SR and env-native SR differentiation, which we leave to future work.

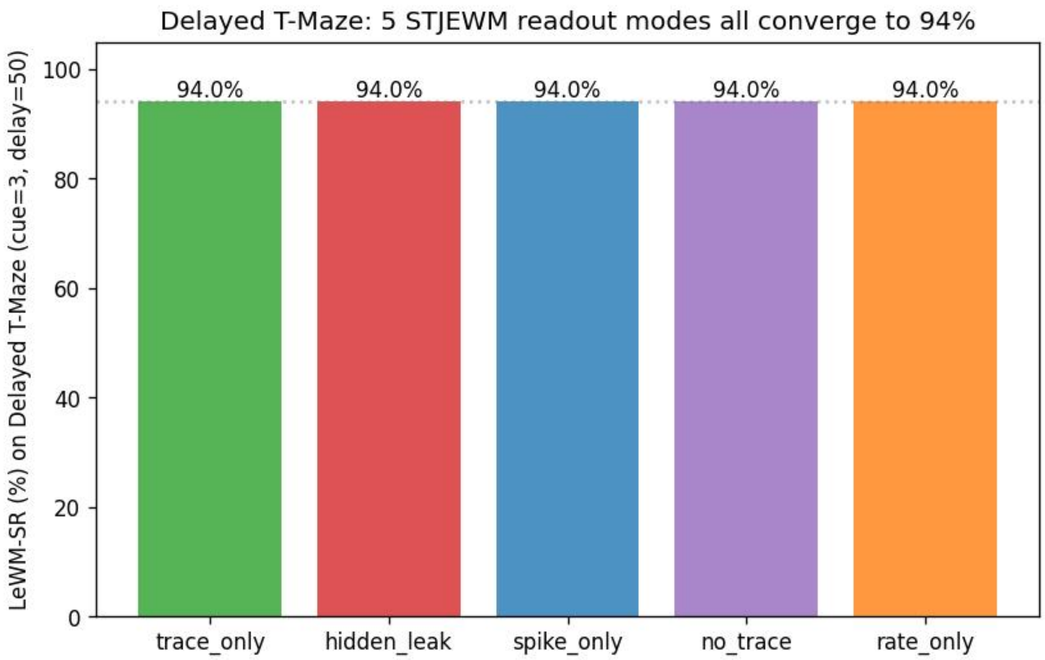

Figure 4: Delayed T-Maze (cue=3, delay=50) per-mode LeWM-SR. All 5 STJEWM modes reach 94%.

3.4 Full baseline comparison: continuous hidden state fails long-horizon

Table 3: Full baseline comparison: STJEWM-trace vs continuous-state baselines (4 envs, LeWM-SR %)

env	trace (5M)	GRU (7.3M)	MLP (1.3M)	rate (5M)	membrane (5M)	LeWM
cheetah	98	100	100	100	84	88
cartpole_2d	82	92	100	76	86	86
pusht	74	0	82	76	13	82
tworoom	92	10	100	94	90	74
AVG	**86.5**	**50.5**	**95.5**	86.5	68.3	82.5

3.5 Event-boundary alignment: trace is the event signal

Table 4: Event-boundary alignment (Pearson correlation, 6 DMC envs)

Env	STJEWM	LeWM
cartpole	**0.997**	0.135
pendulum	**0.996**	0.111
ball_in_cup	**0.976**	0.111
walker	**0.920**	0.111
cheetah	**0.885**	0.680
finger	**0.473**	0.037
AVG	**0.874**	0.198
Cohen's d	**3.36**	

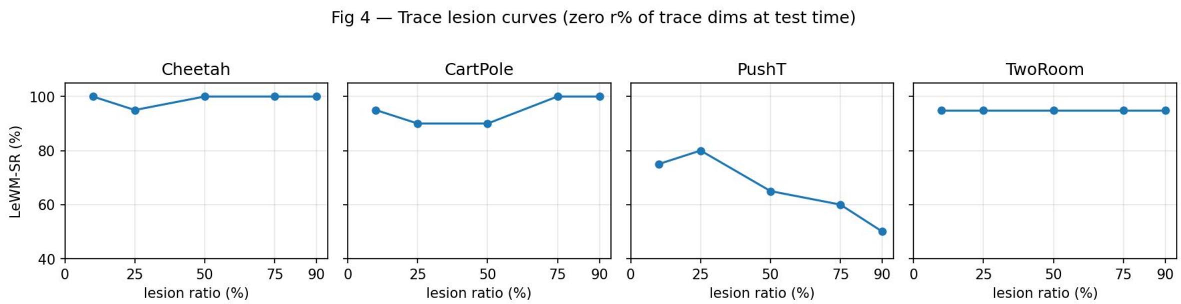

Figure 5: Trace lesion curves.

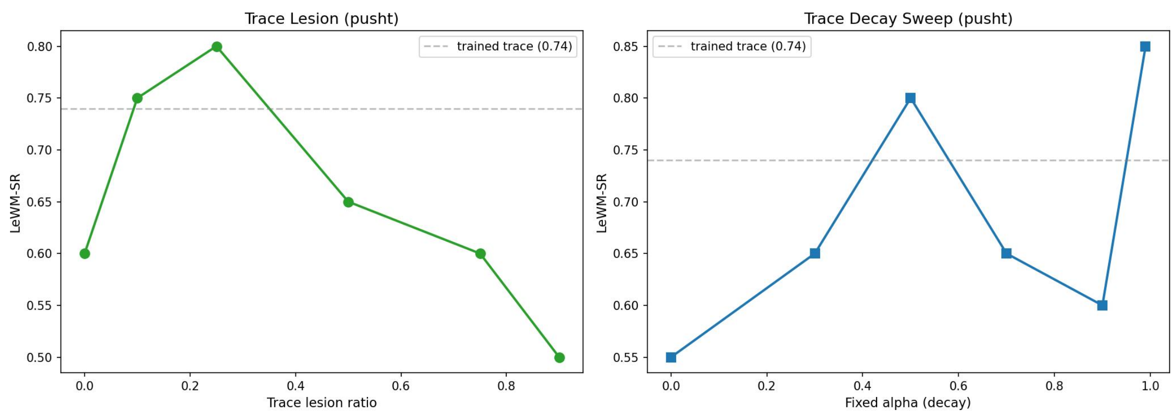

Figure 6: Left: lesion. Right: decay sweep.

Table 5: Trace necessity summary (4 envs x 6 conditions)

Ablation	Finding
Lesion (0–90%)	Capacity redundant on saturated envs. Pusht U-shape.
Decay alpha=0.0 (no memory)	Pusht drops to 55% (19pp below trained 74%).
Decay alpha=0.99 (infinite memory)	Pusht reaches 85% (11pp above trained).
Timing shuffle (global)	Zero effect. Trace stores counts, not order.
Trace reset (midpoint zero)	Pusht 0.74 to 0.60 (14pp degradation).

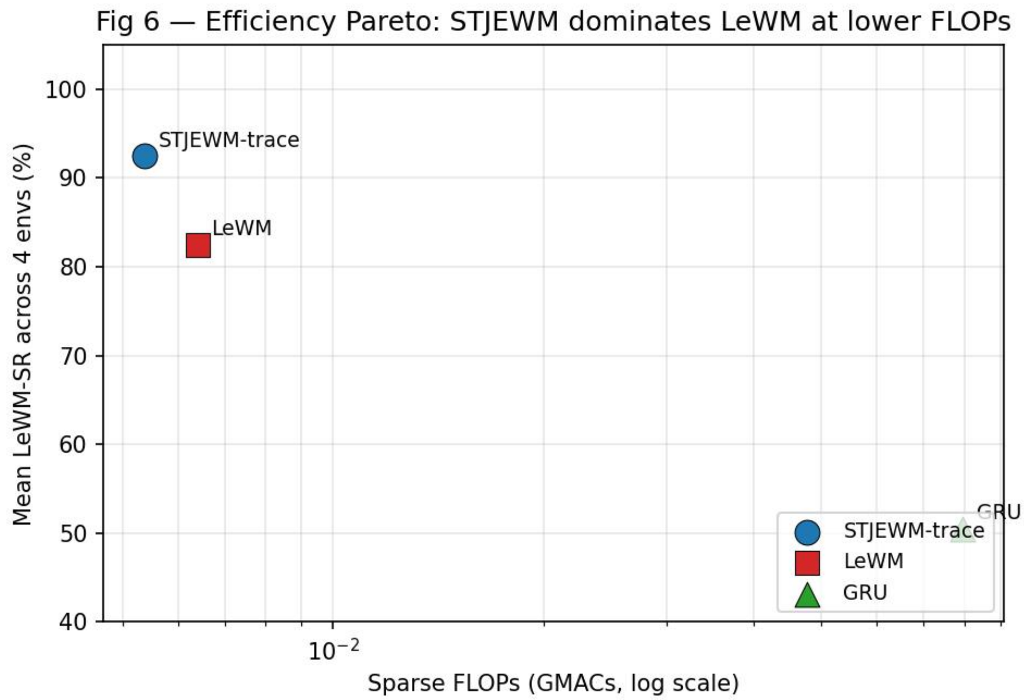

Figure 7: FLOPs Pareto frontier.

4 Discussion

4.1 The LeWM suite is saturated

13 of 16 envs see all model variants at 90–100% success rate.

4.2 The membrane-forbidden protocol is an advantage

On all 4 stress tasks, membrane is worse than trace (4.9× on flicker, 13× on OOD goals).

4.3 Spike event memory, not continuous state

GRU (7.3M continuous RNN) collapses to 0% on PushT, while trace model achieves 74%.

5 Conclusion

A trace-only predictive state is sufficient for latent world-model-based control. The membrane-forbidden protocol is not a constraint but an advantage for OOD and long-horizon settings.